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|--|---|-----------------------------|
| 3 | <p>(a) <u>point</u> at which (whole) weight (of body) (allow mass for weight)
appears / seems to act ... (for mass need 'appears to be concentrated')</p> | <p>M1
A1 [2]</p> |
| <p>(b) (i) point C shown at centre of rectangle ± 5 mm</p> | | <p>B1 [1]</p> |
| <p>(ii) arrow vertically downwards, from C with arrow starting from the same margin of error as in (b)(i)</p> | | <p>B1 [1]</p> |
| <p>(c) (i) reaction / upwards / supporting / normal reaction force
friction
force(s) at the rod</p> | | <p>M1
M1
A1 [3]</p> |
| <p>(ii) comes to rest with (line of action of) weight acting through rod
allow C vertically below the rod
so that <u>weight</u> does not have a moment about the pivot / rod</p> | | <p>B1
B1 [2]</p> |

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4 (a) energy = average force $\times$ extension

P1